Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 2 Quiz 1 - Kinematics in Two Dimensions

Mock Exam E · Form 2: Constraint analysis

Multiple Choice Questions: Select the best answer. (1 point each)

1. A student resolves a 18 m/s velocity directed 30° north of west. East is +x and north is +y. What is its x component?

(A) -15.588 m/s

(B) 15.588 m/s

(C) -31.177 m/s

(D) -7.794 m/s

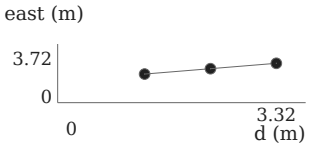
2. A map route starts at east coordinate 2 m and runs at 60° north of east. Rotate the route to 120° north of east at the same starting point. What is the new intercept minus the original intercept?

(A) 0

(B) 0.105

(C) -0.21

(D) 0.315



3. A ball is launched horizontally at 7 m/s from a table 4.9 m high. Ignore drag. Find the velocity angle below horizontal at landing. Use g = 9.8 m/s².

(A) 108.925 °

(B) 27.231 °

(C) 54.462 °

(D) -54.462 °

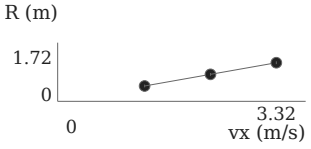
4. A ball launches horizontally from a 1.25 m table. Use g=10 m/s² and neglect drag. vx (m/s) -> R (m): (1.32, 0.66); (2.32, 1.16); (3.32, 1.66). With vertical error bounded by 0.06 and an exact model, what is the error bound on an inferred X?

(A) 0.37

(B) 0.245

(C) -0.005

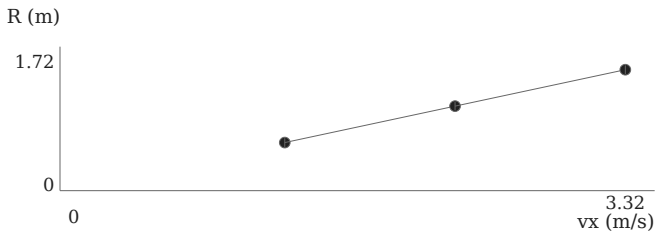
(D) 0.12



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. A ball launches horizontally from a 1.25 m table. Use g=10 m/s² and neglect drag. vx (m/s) -> R (m): (1.32, 0.66); (2.32, 1.16); (3.32, 1.66).

a. For the horizontal-launch target setup, label the measured quantities and their reference directions or origins. **(1 pt)**



b. Quadruple the table height. Predict the new slope and intercept and explain physically what changed. **(3 pt)**

6. A ball leaves a 4 m ledge horizontally at 5 m/s. Use g=10 m/s² and consider airborne frames only. x² (m²) -> y (m): (1.32, 3.736); (2.32, 3.536); (3.32, 3.336).

a. Build the model for airborne video frames from physics and predict both slope and intercept. **(2 pt)**

b. Double horizontal launch speed at the same height. Predict the new slope and intercept and explain physically what changed. **(2 pt)**

7. A map route starts at east coordinate 2 m and runs at 60° north of east. d (m) -> east (m): (1.32, 2.66); (2.32, 3.16); (3.32, 3.66). Build the model for eastward projection from physics and predict both slope and intercept. **(3 pt)**